

Task 1 (2 marks)
Intuitive design of data cubes

(1) 0.3 mark

Consider the following description of a sample database domain.

A transportation company collects and stores data about the trips performed by its drivers. The company saves the data in a multidimensional data cube.

The company would like to use the data to compute an average fuel consumption per day, per month and per year, an average length of trips measured in kilometers per month and per year, and an average length of trips measured in hours.

Use a short explanation of a database domain given above to find a multi-dimensional data cube that should be implemented. In your specification of a data cube, list the facts, the measures, the names of dimensions and the hierarchies.

Facts:

Trips performed by company drivers

Measures:

Fuel consumption per trip
Length of trip in kilometres
Length of trip in hours

Dimensions:

Date/Time

Hierarchies

Year ---< Month ---< Day ---< Hour

(2) 0.7 mark

Consider the following description of a sample database domain.

A transportation company collects and stores data about the trips performed by its drivers. The company saves the data in a multidimensional data cube.

The company would like to use the data to compute an average fuel consumption per day, per month and per year, an average length of trips measured in kilometers per month and per year, and an average length of trips measured in hours.

Following the successful implementation of the first stage, the company decided to extend the system. In the second stage, the company aims to compute the total number of trips per driver, per day, per month and per year and the total number of driving days per

driver and year. Drivers are uniquely identified by their driving license number, first name, and last name.

Use a short explanation of a database domain given above to find a multi-dimensional data cube that should be implemented. In your specification of a data cube, list the facts, the measures, the names of dimensions and the hierarchies.

Facts:

Trips performed by company drivers

Measures:

Fuel consumption per trip

Length of trip in kilometres

Length of trip in hours

Dimensions:

Date/Time

Driver

Hierarchies

Year ---< Month ---< Day ---< Hour

(3) 1.0 mark

Consider the following description of a sample database domain.

A transportation company collects and stores data about the trips performed by its drivers. The company saves the data in a multidimensional data cube.

The company would like to use the data to compute an average fuel consumption per day, per month and per year, an average length of trips measured in kilometers per month and per year, and an average length of trips measured in hours.

Following the successful implementation of the first stage, the company decided to extend the system. In the second stage, the company aims to compute the total number of trips per driver, per day, per month and per year and the total number of driving days per driver and year. Drivers are uniquely identified by their driving license number, first name, and last name.

Following the successful implementation of the second stage, the company has decided to further extend the system. This third stage focuses on computing the total trips performed by different vehicle types. The company's fleet includes cars, buses, and trucks. Each vehicle is generally described by its registration number, make, model, and fuel consumption. Additionally, buses are characterized by their total available seats, while trucks are described by their total capacity.

Use a short explanation of a database domain given above to find a multi-dimensional data cube that should be implemented. In your specification of a data cube, list the facts, the measures, the names of dimensions and the hierarchies.

Facts:

Trips performed by company drivers

Measures:

Fuel consumption per trip

Length of trip in kilometres

Length of trip in hours

Dimensions:

Date/Time

Driver

Vehicle

Hierarchies

Year ---< Month ---< Day ---< Hour

Vehicle-type --< Vehicle
